APS110H1f2026 - September 21, 2026

September 21, 2026 9:55 AM

- Purity/Purities
- Gorilla Glass
- Properties of Matter(s) - today (and also for this week upcoming)
  - Bonding Strengths
  - Density/Mass

Purity/Purities

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- purity SiO_2 100% - 99.9% - 99.99%

- impurity/impurities SiO_2 95% GeO_2 5%

4+ 2- 4+ 2-

silicon Germanium

Silicon dioxide Germanium dioxide

Gorilla Glass

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pm = picometre nm = nano meter.

210 pm = 0.21 nm

O - Si - O bondings

152 pm

O - Al - O - Al - O bondings

184 pm

not needed for today.

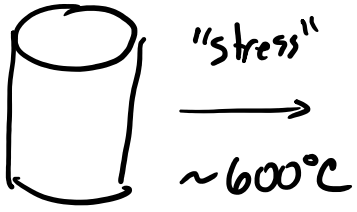
Na - O - Na or K - O - K

227 pm 275 pm

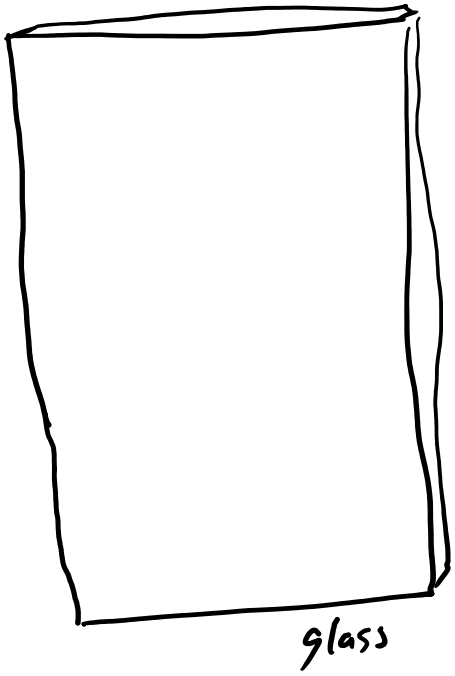
Si^{4+}	SiO_2	60-75%
Al^{3+}	Al_2O_3	5-15%
Na^{1+} / K^{1+}	Na_2O or K_2O	8-18%
	sodium	potassium

Stabilizers

↑ low-purity



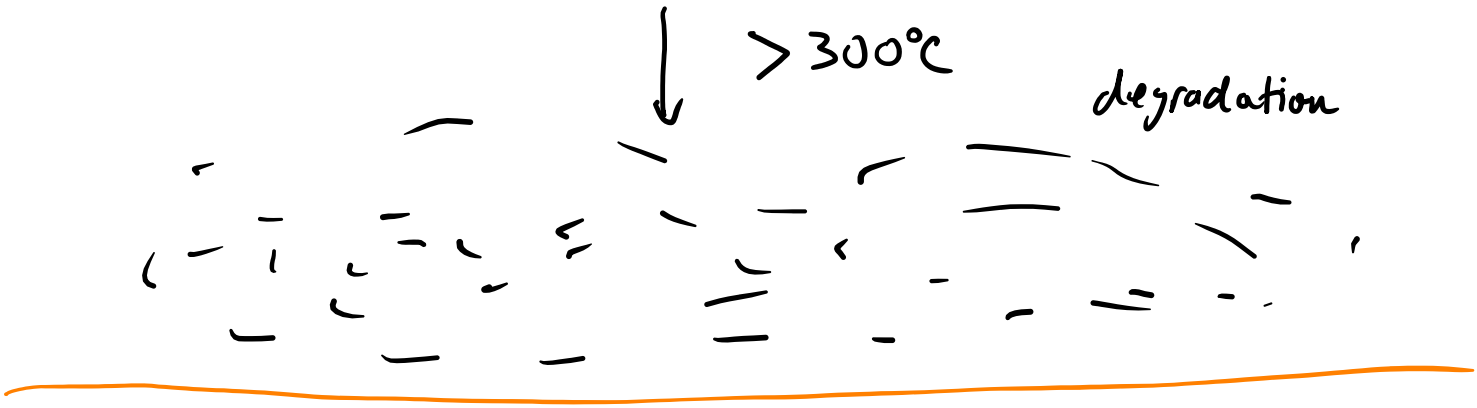
Starting chemicals ceramics



Gorilla Glass

- of metals + ceramics - sometimes takes high temperature metals
- but plastics/polymer.

- If above ~300 temp, then degradation of the polymers/plastics…


> 300°C
degradation

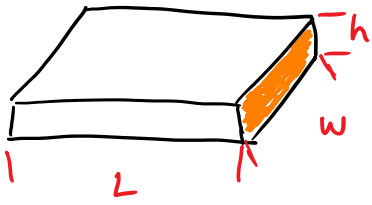
Properties of Matter(s)

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mm - millimetre

cm - ceftimetre.

$A = h + w$



$h - 4.0\ mm = 0.4\ cm$

$w - 154\ mm = 15.4\ cm$

$L - 406\ mm = 40.62\ cm.$

Just as examples re numbers for today…



ρ - density

$mass = m \Rightarrow m = A \times L \times \rho$

$m = h \times w \times L \times \rho$

$density = \rho \Rightarrow \rho = \frac{m}{A \times L} = \frac{m}{h \times w \times L}$